# CURRICULUM VITAE

## Alexios Voulimeneas

Nationality: Greek
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Languages: Greek (native), English (fluent), German (basic)

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## Research Interests

**Cyber Security:** software diversity, memory safety, sandboxing, compartmentalization, exploits, defenses

**Operating Systems:** kernel development, networking, debugging, virtualization, application monitoring, heterogeneous computing, fault tolerance, reliability, record/replay

**Software Protection:** anti tampering, integrity checking, reverse engineering

## Appointments

**Assistant Professor** 2023 – Present
TU Delft
Proud member of the Cybersecurity (CYS) section in Computer Science, Department of Intelligent Systems, Faculty of Electrical Engineering, Mathematics, and Computer Science. Tenured since March 15th 2025.

**Visitor Researcher** 05/2025 –10/2025
European Space Agency

**Postdoctoral Scholar** 2020 – 2023
KU Leuven
I worked with Professor Stijn Volckaert in the DistriNet research group at KU Leuven's Technology Campus in Ghent, Belgium.

**Graduate Research Assistant** 2015 – 2020
University of California, Irvine
I worked with Professor Michael Franz in the Secure Systems Lab at the Donald Bren School of Information and Computer Sciences.

**Visiting Scholar** Fall 2019
KU Leuven

**Software Engineering Intern** Summer 2019
Apple Inc.

**Research Assistant Intern** Summer 2017
Oracle Labs

**Hellenic Army** (mandatory military service) 11/2014 – 08/2015
Research and Informatics Directorate

**Undergraduate and Graduate Researcher** 2011 – 2014
Mobile Multimedia Laboratory/AUEB

**C and JAVA Software Engineer Intern** Summer 2012
NCSR Demokritos

# Education

**PhD in Computer Science** 2020
University of California, Irvine
**Advisor:** Professor Michael Franz
**Thesis Topic:** Building the Next Generation of Security Focused NVX Systems: Overcoming Limitations of N-Variant Execution

**MSc in Computer Science** 2017
University of California, Irvine

**Ptychio in Informatics (4-year degree)** 2012
Athens University of Economics and Business (AUEB)/Department of Informatics, Athens
**Advisor:** Professor George Xylomenos
**Thesis Topic:** Towards an Error Control Scheme for a Publish/Subscribe Network

# Awards, Grants, and Distinctions

2024: ACSAC Distinguished Paper Award with Artifact
2023: CCS Top Reviewer Award
2023: ASIA CCS Best Reviewer Award
2022: Runner-up for the 2022 CNIL-Inria Award for Privacy Protection
2022: EuroSys Distinguished Reviewer Artifact Award
2016: ACM CCS Student Travel Grant
2016: IEEE S&P Student Travel Grant
2015: ICS Dean's Award, University California, Irvine
2013: Scholarship for Graduate Studies, Latsis Foundation
2013: Scholarship for Graduate Studies, Foundation for Education and European Culture
2012: Valedictorian Graduate, Athens University of Economics and Business

# Publications

**IBTpoline: Flexible, Robust, and Intel IBT-compliant System Call Hooking**
Alexander J. Gaidis, Ioulios Patmanidis, Isabelle Shapiro, Georgios Portokalidis, Alexios Voulimeneas, and Vasileios P. Kemerlis.
To appear in Proceedings of the SIGOPS Annual Technical Conference (ATC), Hong Kong, Hong Kong, November 2026.
***[13.9% acceptance rate]***

**Clair Obscur: The Light and Shadow of System Call Interposition – From Pitfalls to Solutions with K23**
Jesús María Gómez, Vissarion Moutafis, Antreas Dionysiou, Fernando Kuipers, Georgios Smaragdakis, Bart Coppens, and Alexios Voulimeneas.
*In ACM/IFIP International Middleware Conference (Middleware 2025).*
***[21% acceptance rate (not including rollover papers)]***

**Divide and Conquer: Introducing Partial Multi-Variant Execution.** J. Vinck, A. Jacobs, **A. Voulimeneas**, and S. Volckaert. *In IEEE European Symposium on Security and Privacy (Euro S&P 2025).*

**Moneta: Ex-Vivo GPU Driver Fuzzing by Recalling In-Vivo Execution States.** J. Jung, J. Jang, Y. Jo, J. Vinck, **A. Voulimeneas**, S. Volckaert, and D. Song. *In Network and Distributed System Security Symposium (NDSS 2025).*
***[211 papers accepted out of 1311 submissions = 16.1%]***

**I'll Be There for You! Perpetual Availability in the A⁸ MVX.** A. Rösti, S. Volckaert, M. Franz, and **A. Voulimeneas.** *In Annual Computer Security Applications Conference (ACSAC 2024).*
***[83 papers accepted out of 381 submissions = 21.8%] [Distinguished Paper with Artifact Award]***

**The Astonishing Evolution of Probabilistic Memory Safety: From Basic Heap-Data Attack Detection towards Fully Survivable Multi-Variant Execution.** A. Rösti, **A. Voulimeneas,** and M. Franz. *In IEEE Security & Privacy 2024.*

**Orbital Shield: Rethinking Satellite Security in the Commercial Off-the-Shelf Era.** N. Yadav, F. Vollmer, A. R. Sadeghi, G. Smaragdakis, and **A. Voulimeneas.** *In IEEE Conference on Security for Space Systems (3S) 2024.*

**System Call Interposition Without Compromise.** A. Jacobs, M. Gülmez, A. Andries, S. Volckaert, and **A. Voulimeneas.** *In IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2024).* ***[42 papers accepted out of 203 submissions = 20%]***

**A run a day won't keep the hacker away: Inference attacks on endpoint privacy zones in fitness tracking social networks.** K. Dhondt, V. L. Pochat, **A. Voulimeneas,** W. Joosen, and S. Volckaert. *In BlackHat Asia 2023*

**A run a day won't keep the hacker away: Inference attacks on endpoint privacy zones in fitness tracking social networks.** K. Dhondt, V. L. Pochat, **A. Voulimeneas,** W. Joosen, and S. Volckaert. *In ACM Conference on Computer and Communications Security (CCS 2022).* ***[218 papers accepted out of 971 submissions = 22.5%]***

**You Shall Not (by)Pass! Towards Secure, and Fast PKU-based Sandboxing. A. Voulimeneas,** J. Vinck, R. Mechelinck, and S. Volckaert. *In European Conference on Computer Systems (EuroSys 2022).* ***[42 papers accepted out of 162 submissions = 25.9%]***

**Sharing is Caring: Secure and Efficient Shared Memory Support for MVEEs.** J. Vinck, B. Abrath, B. Coppens, **A. Voulimeneas,** B. De Sutter, and S. Volckaert. *In European Conference on Computer Systems (EuroSys 2022).* ***[42 papers accepted out of 162 submissions = 25.9%]***

**dMVX: Secure and Efficient Multi-Variant Execution in a Distributed Setting. A. Voulimeneas,** D. Song, P. Larsen, M. Franz, and S. Volckaert. *In European Workshop on Systems Security (EuroSec 2021)*

**Distributed Heterogeneous N-Variant Execution. A. Voulimeneas,** D. Song, F. Parzefall, Y. Na, P. Larsen, M. Franz, and S. Volckaert. *In Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2020).* ***[13 papers accepted out of 45 submissions = 28.9%]***

**Secure and Efficient Application Monitoring and Replication.** S. Volckaert, B. Coppens, **A. Voulimeneas,** A. Homescu, P. Larsen, B. De Sutter, and M. Franz. In *USENIX Annual Technical Conference (ATC 2016).* ***[47 papers accepted out of 266 submissions = 17.6%]***

**A Reliable Multicast Transport Protocol for Information-Centric Networks.** C. Stais, G. Xylomenos, and **A. Voulimeneas**. *In Journal of Network and Computer Applications (JNCA 2014)*

**Towards an Error Control Scheme for a Publish/Subscribe Network**. C. Stais, **A. Voulimeneas**, and G. Xylomenos. *In International Conference on Communications (ICC 2013)*

**Theses**

**Building the Next Generation of Security Focused NVX Systems: Overcoming Limitations of N-Variant Execution.** A. Voulimeneas. PhD Thesis, 2020.

**Towards an Error Control Scheme for a Publish/Subscribe Network**. A. Voulimeneas. BSc Thesis, 2012.

# Funding

HONEY-MON: Cyber Deception via N-Variant Execution — ONR (2021-2024)
I contributed to the conception/brainstorming/writing of this accepted proposal
Michael Franz (sole PI)
Award Amount: +/- 1.65M $

TU Delft Starting Package — TU Delft (2023)

Funding for one PhD student for my appointment at TU Delft
Alexios Voulimeneas
Award Amount: +/- 375K €

# Professional Scientific Activities

## Service

(CCS) ACM Conference on Computer and Communications Security, Program Committee [2023, 2024, 2026, 2027]
(ORCA) Open Robust Compartmentalization Alliance, Technical Steering Committee [2026]
(EuroSys) European Conference on Computer Systems, Proceedings Chair [2025, 2027]
(EuroSys) European Conference on Computer Systems, Program Committee [2023, 2027]
(ACM REP) ACM Conference for Reproducibility and Replicability, General Chair [2026]
(RAID) International Symposium on Research in Attacks, Intrusions and Defenses, Program Committee [2026]
(ISSN 2078-2489) Information, Reviewer, [2025]
(CCS) ACM Conference on Computer and Communications Security, Treasurer [2026]
(HotOS) SIGOPS ACM Workshop on Hot Topics in Operating Systems, Proceedings Chair [2027]
(SOSP) ACM Symposium on Operating Systems Principles, Artifact Evaluation Committee Chair [2026]
(ACSAC) Annual Computer Security Applications Conference, Program Committee [2025, 2026]
(SYSTOR) ACM International Systems and Storage Conference, Program Committee [2025]
(ACNS) International Conference on Applied Cryptography and Network Security, Program Committee [2025]
(S&P) IEEE Symposium on Security and Privacy, Program Committee [2025, 2026, 2027]
(ISC) Information Security Conference, Program Committee [2024, 2025]
(EuroSec) European Workshop on System Security, Program Committee [2024, 2025, 2026]
(USENIX Security) USENIX Security Symposium, Program Committee [2024, 2025, 2026, 2027]
(DIMVA) Conference on Detection of Intrusions and Malware & Vulnerability Assessment, Program Committee [2024]
(Middleware) ACM/USENIX/IFIP International Middleware Conference, Program Committee [2023, 2025]
(USENIX Security) USENIX Security Symposium, Artifact Evaluation Publication Chair [2023]
(ASIACCS) ACM ASIA Conference on Computer and Communications Security, Program Committee [2023, 2024]
(ESORICS) European Symposium on Research in Computer Security, Program Committee [2022, 2023]
(PLDI) ACM SIGPLAN Conference on Programming Language Design and Implementation, Artifact Evaluation Committee [2022]
(EuroSys) European Conference on Computer Systems, Artifact Evaluation Committee [2022]
(USENIX Security) USENIX Security Symposium, Artifact Evaluation Committee [2022]
(EuroSys) European Conference on Computer Systems, External Reviewer [2021, 2022]
(ROOTS) Reversing and Offensive-oriented Trends Symposium, Program Committee [2020, 2021, 2022]
(OSDI) USENIX Symposium on Operating Systems Design and Implementation, Artifact Evaluation Committee [2020]
(S&P) IEEE Symposium on Security and Privacy, Student Program Committee [2018]

## Conferences Attended (+ = I gave a presentation)

(ASPLOS) International Conference on Architectural Support for Programming Languages and Operating Systems [Attended in 2025]
(ACSAC) Annual Computer Security Applications Conference [Attended in 2024]
(ESORICS) European Symposium on Research in Computer Security [Attended in 2023]
(EuroSys) European Conference on Computer Systems [Attended in 2022+, 2025]
(NDSS) Network and Distributed Systems Security Symposium [Attended in 2016, 2017, 2018, 2025]
(S&P) IEEE Symposium on Security and Privacy [Attended in 2016]
(CCS) ACM Conference on Computer and Communications Security [Attended in 2016]
(DIMVA) Conference on Detection of Intrusions and Malware & Vulnerability Assessment [Attended in 2020+, 2022+]
(EuroSec) European Workshop on Systems Security [Attended in 2020+]

## Talks

**Interpose Me If You Can – Pitfalls and Solutions in System Call Interposition When Performance and Usability Matter**
Invited Talk, National and Kapodistrian University of Athens, Greece, Athens, February 2026

**Interpose Me If You Can – Pitfalls and Solutions in System Call Interposition When Performance and Usability Matter**
Invited Talk, ORCA (Open Robust Compartmentalization Alliance under the Linux Foundation), Online, November 2025

**Interpose Me If You Can – Pitfalls and Solutions in System Call Interposition When Performance and Usability Matter**
Invited Talk, University of Lisbon & INESC-ID Lisbon, Lisbon, Portugal, October 2025

**You Shall Not (by)Pass! Practical, Secure, and Fast PKU-based Sandboxing**
Invited Talk, ORCA (Open Robust Compartmentalization Alliance under the Linux Foundation), Online, July 2025

**Building Secure and Reliable Systems – A Systems Approach**
Invited Talk, Ghent University, Ghent, Belgium April 2025

**Building Secure and Reliable Systems – A Systems Approach**
Invited Talk, TU Delft, PL Group, Delft, Netherlands, November 2024

**Building Secure and Reliable Systems – A Systems Approach**
Invited Talk, IMDEA Software Institute, Madrid, Spain, September 2024

**Tales of Memory-Error Exploits and Defenses**
Invited Talk, TU Delft, Delft, Netherlands, March 2023

**Tales of Memory-Error Exploits and Defenses**
Invited Talk, AUEB, Athens, Greece, December 2022

**You Shall Not (by)Pass! Practical, Secure, and Fast PKU-based Sandboxing**
Invited Talk, Telecom SudParis/IP Paris, Online, December 2022

**You Shall Not (by)Pass! Practical, Secure, and Fast PKU-based Sandboxing**
Invited Talk, Intel Labs, Online, May 2022

**Distributed Heterogeneous N-Variant Execution**
Invited Talk, DRADS DistriNet Workshop, Leuven, Belgium, July 2021

**Redundant Execution and Lightweight Monitoring for Security and Performance**
Invited Talk, Mobile Multimedia Laboratory (MMlab), AUEB, Athens, Greece, June 2019

**Industry and Graduate Studies Opportunities in Greece, Europe, and USA**
Invited Talk, AUEB, Athens, Greece [2015, 2016, 2017, 2019, 2021, 2022]

# Teaching Experience

**Course Developer & Instructor** 2024 – Present
Systems Security
TU Delft

**Course Developer & Instructor** 2024 – Present
Computer Security
TU Delft

**Teaching Assistant** Winter 2017
ICS 6B: Boolean Algebra and Logic
University of California, Irvine

**Reader** Spring 2016
ICS 46: Data Structure Implementation and Analysis
University of California, Irvine

**Lab Assistant** Summer 2010
Computational Mathematics
Athens University of Economics and Business

# Media Coverage

**Article at the Security Management Magazine about our recent ACSAC 2024 paper** July 2025
Netherlands

# Mentoring, Training, and Advancement of Young Scientists

**Postdoc Mentor:**

**ongoing:**

Antreas Dionysiou (2025-Present, TU Delft, Marie Skłodowska-Curie Postdoctoral Fellow Co-Hosting with Georgios Smaragdakis)

**PhD Research and Dissertation Mentor:**

**ongoing:**

Vissarion Moutafis (2025-Present, TU Delft)

Jesús María Gómez Moreno (2024-Present, TU Delft)

**External PhD Thesis Reviewer/Evaluator**

Alessandro Sanna (2025, University of Cagliari), "Under The Surface: Analysing Unconventional Malware and Unveiling its Secrets"

**MSc Thesis Mentor:**

**ongoing:**

Camiel Vletter (2025-Present, TU Delft)

**completed:**

Dimitris Ntatsis (MSc 2026, TU Delft) “Validating Rust Check Safety Using Binary Analysis”

Daniel Péter (MSc 2026, TU Delft) “Metal-fuzz: A Hardware-in-the-Loop Fuzzing Framework for Space Hypervisors”

Ioulios Patmanidis (MSc 2026, TU Delft) “Secure System Call Interposition with PKU-Based Memory Isolation”

Matyáš Kollert (MSc 2026, TU Delft) “Designing Secure Boot for Space Missions”

Einar de Gruyl (MSc 2026, TU Delft) “Proving Compilation of the Java Memory Model”

Yigit Colakoglu (MSc 2026, TU Delft), “Thwarting Code-Reuse Attacks Through Temporal Permission Tightening in User Space”

Ali Kahawati (MSc 2025, TU Delft), “Enhancing Syscall Interposition Safety through Rust - A Memory-Safe Implementation of the Lazypoline Framework”

Vissarion Moutafis (MSc 2025, TU Delft), “MORA: Hunting Space Bugs in your Sleep”

Wouter Jehee (MSc 2024, TU Delft), “WALL-EYE: Taking a look at CubeSat security - Security analysis of CubeSats on a physical testbed”

Michael Poker (MSc 2022, TU Braunschweig), “Techniques for Fast-Forwarding Execution State to Synchronize Software Instances in an MVEE”

Adriaan Jacobs (MSc 2021, KU Leuven), “Combating address-sensitive behavior in MVEEs”

***Current as October 26, 2026***